

IDM Simulation Study - Acceleration Comparison

Selected trajectory pair: 1 | Time step: 0.1 s | Samples: 841

IDM parameters: $v_0=30$ m/s, $s_0=2$ m, $T=1.5$ s, $a=1.0$ m/s², $b=1.5$ m/s², $\delta=4$

This plot compares leader acceleration, observed follower acceleration, and IDM-simulated follower acceleration from the NGSIM dataset. Follower motion was initialized from the recorded first state and updated over time using Euler's method.

Acceleration RMSE (real vs IDM simulated follower): 1.984 m/s²

Acceleration vs Time

